Deliverable 1: The Artifact

What was built, and what it looks like running

Mills-Operation · AI Reliability Copilot for Hot Mill Operations

The Artifact

This is a working, end to end reliability console for a hot steel mill line. It is not a mockup, every screen below is a real screenshot of the running app, and every number on it comes from a trained model scoring real (synthetic) sensor data.

What it does

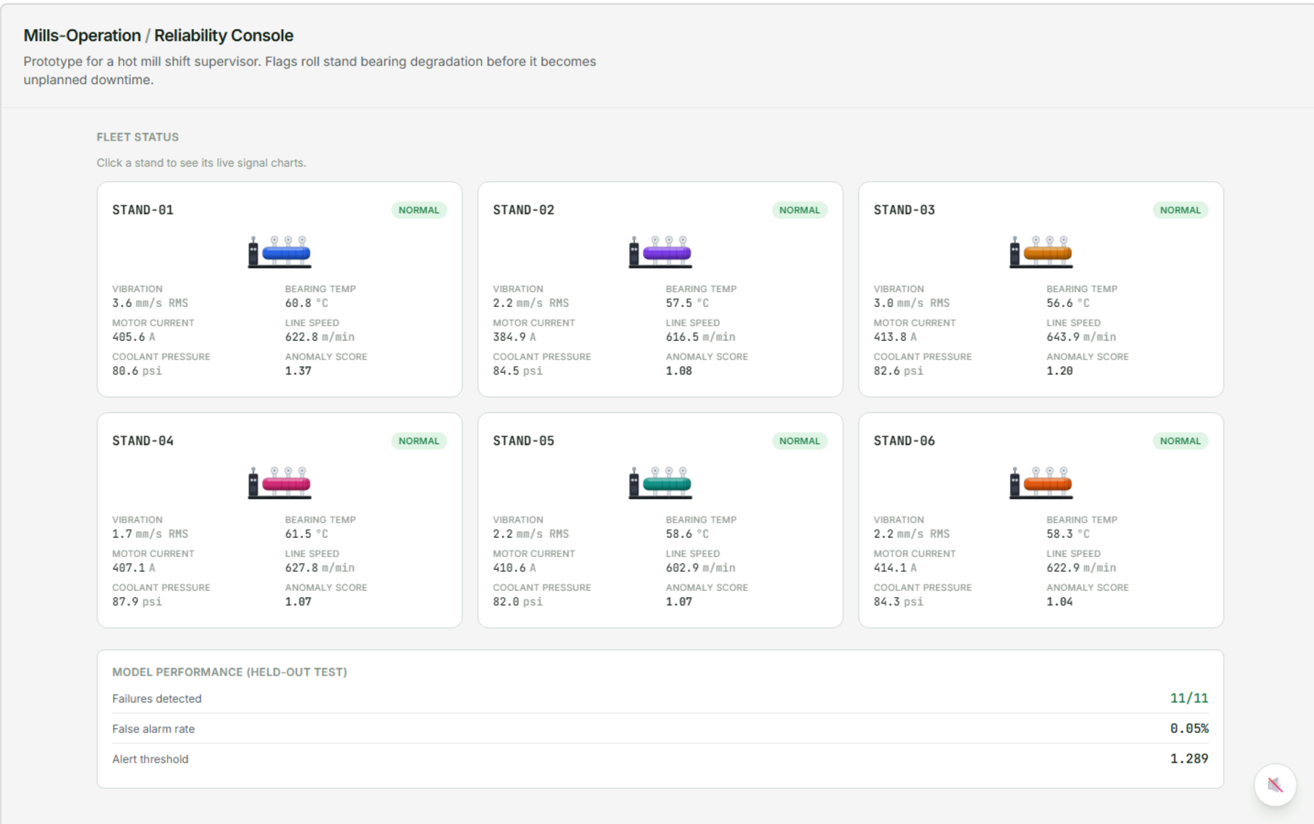
A shift supervisor gets three things in one console:

1. **Live Monitor**, 6 roll stands ticking once every real minute, scored live, with a date range filter (today, past week, last month, or a custom range) that reaches back through the training data.
2. **Degradation Simulator**, a virtual stand the supervisor can push toward failure by hand, to see how the model actually responds, minute by minute.
3. **Fleet Copilot**, a real chat that answers questions about the fleet using 6 read only tool calls into live data, never invented numbers.

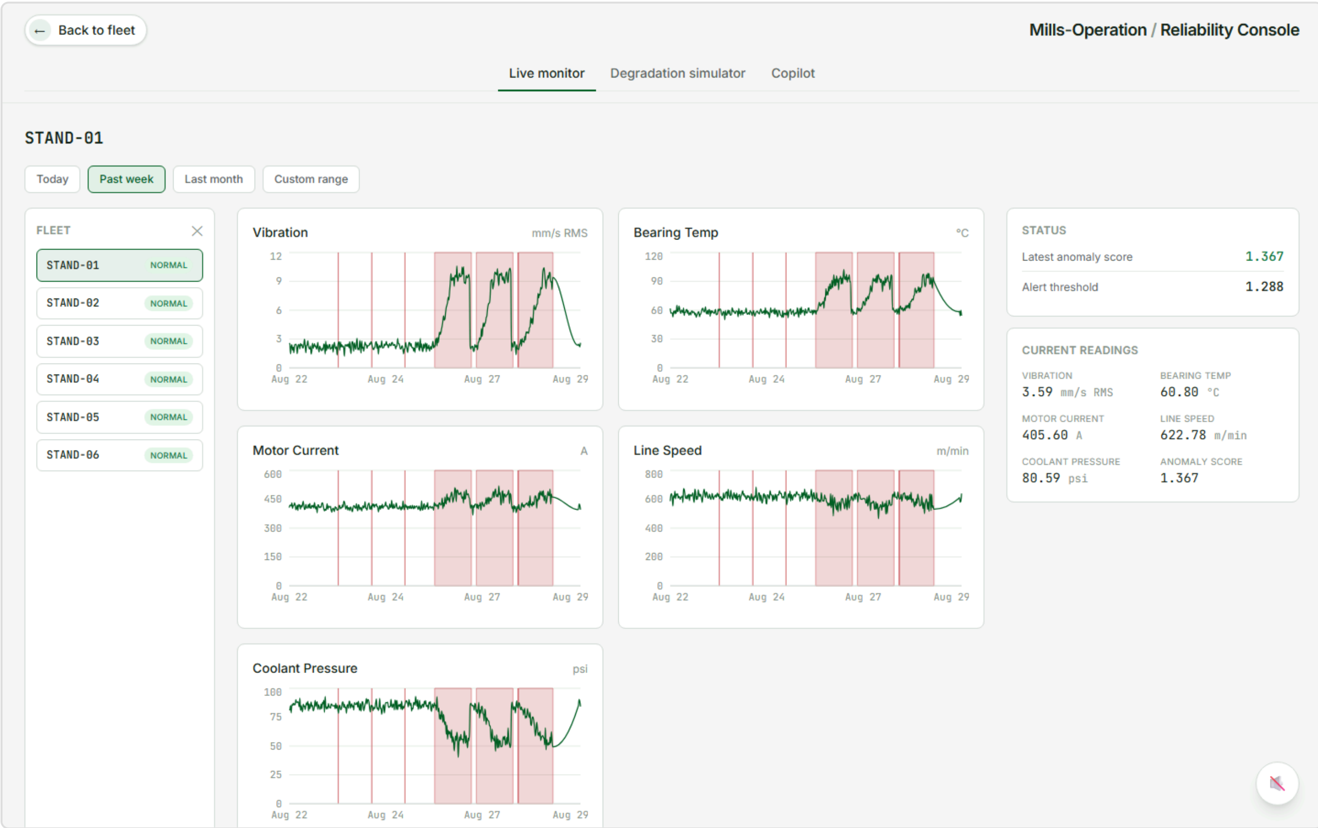
How to see it running

The code and a full README with setup steps are in the accompanying code deliverable. Clone the repository, run the two setup commands, start the backend and frontend, and the console is live at localhost.

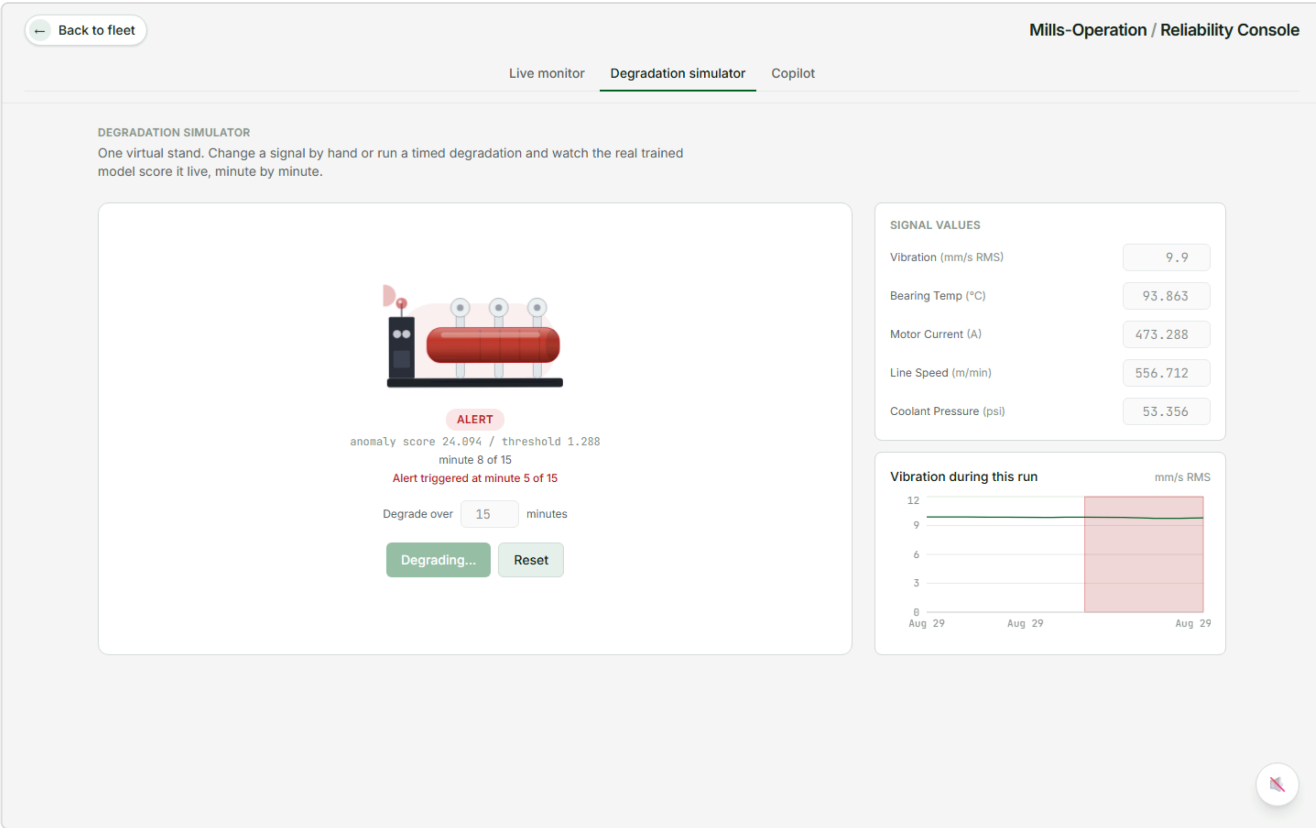
Screenshots



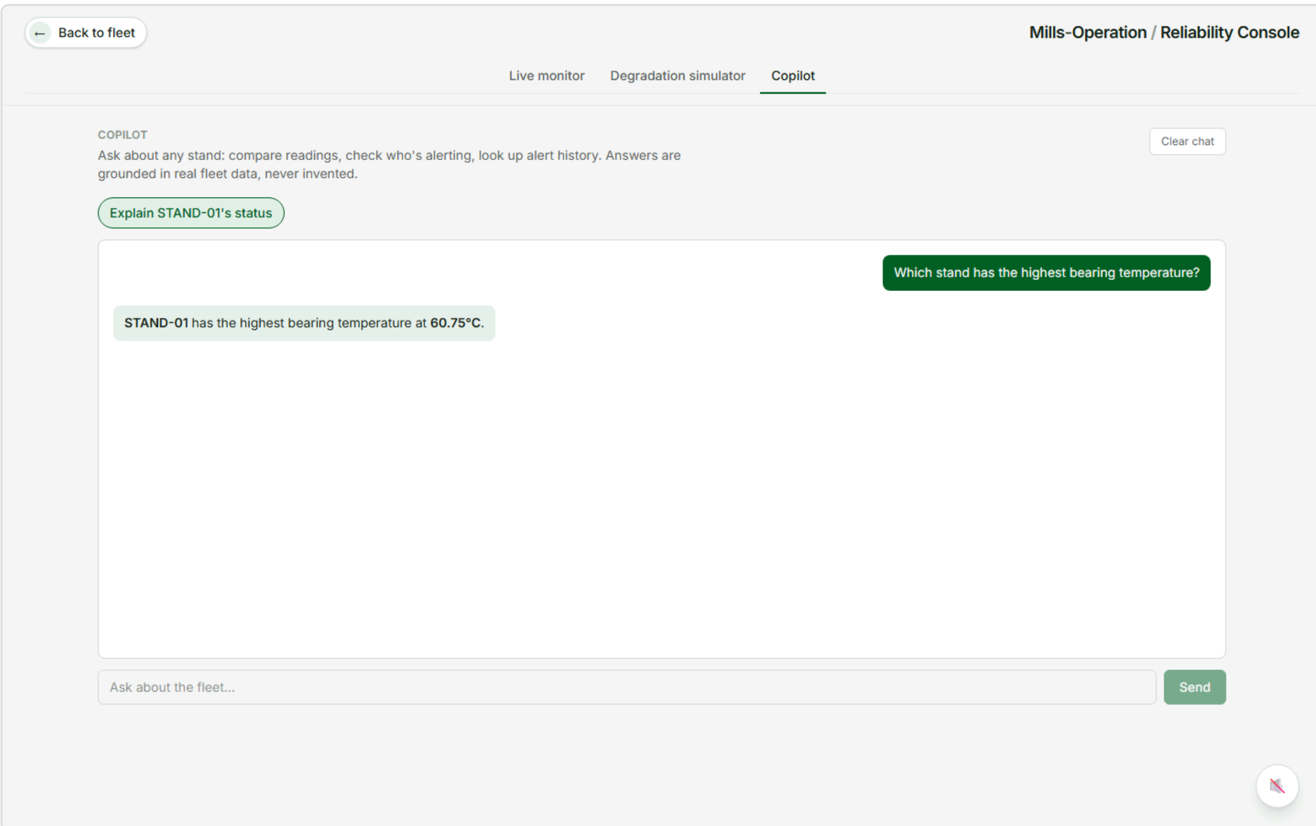
The fleet landing page. All 6 stands, live sensor values, and the model's held out test performance (11/11 failures detected, 0.05% false alarm rate).



A single stand's signal history over the past week, with real detected alert windows shaded in.



The degradation simulator, live. Vibration pushed toward failure by hand, and the real trained model has already flagged the alert.



The fleet copilot answering a free text question, grounded in a live tool call, not a canned response.